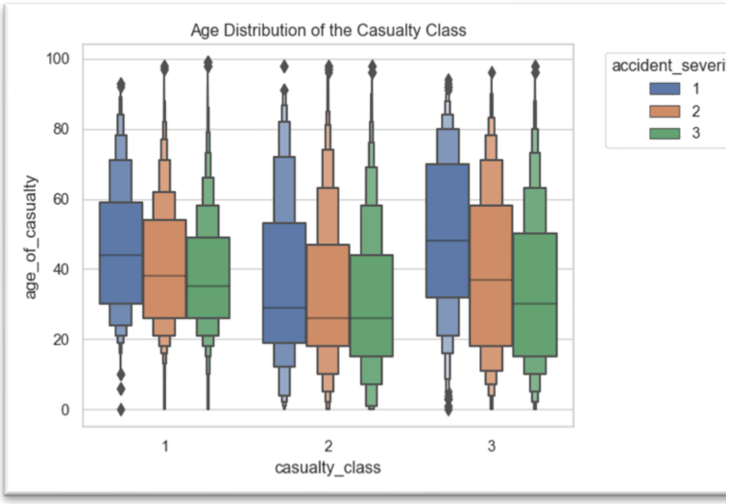
| Project – Analyzing Accident Data | Dataset: accident_data_v1.0.0 2023.db |
|---|---|

**Introduction.**

The dataset used in this report is accident_data_v1.0.0 2023.db, a real-world dataset of road traffic accidents in Great Britain, with focus on year 2020. The first task is to analyse the data and advise government agencies on how to improve road safety. Second task is to create a model that would predict such accidents and the injuries that they incur.

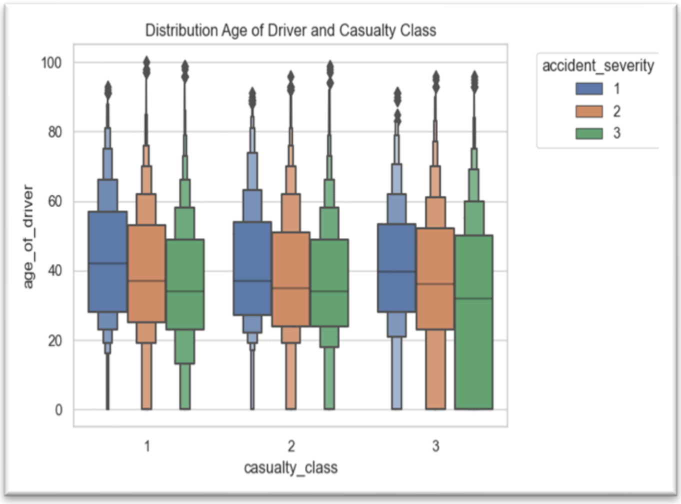
**Exploratory Data Analysis: Analysing Road Traffic Accidents by Casualty Factors**

### I. Age Distribution of the Casualties




- This boxenplot above shows the median age for casualty-class 1 involved in fatal accidents to be 45 years, 30 years for casualty-class 2 and 50 years for casualty-class 3 .
- For serious accidents, it is 38 years for casualty-class 1  27 years for class 2 and 37 years for casualty-class 3

### II Age distribution of drivers involved in accidents

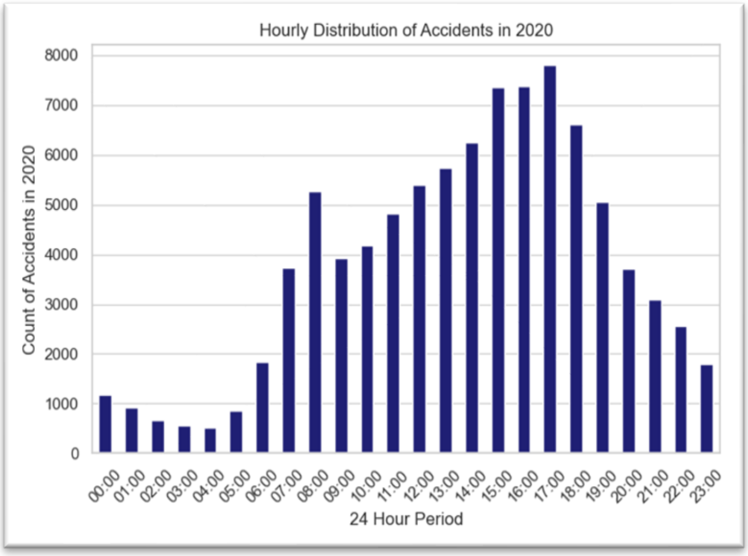



- The boxenplot above demonstrates the median age for drivers involved fatal accidents to be 45, 35 and 40 for casualty class 1, 2, and 3 respectively.
- It is observed from the plot above indicated that the median age of drivers involved in slight accidents with casualty-class 3 to be 30. 75% of these drivers are less than 50 years of age, 50% less than 30 years. And 25% of these drivers were documented as -1 (unknown).

**Task No. 1**
Significant hours of the day, and days of the week, on which accidents occur

### Significant hours of the day in which accidents occur

To examine the hour(s) of the day in which most accidents occur, a temporarily feature (n_time) was created as a datetime64[ns]object from the time feature, to enable analysis of accidents that occurred in each hour of the day.

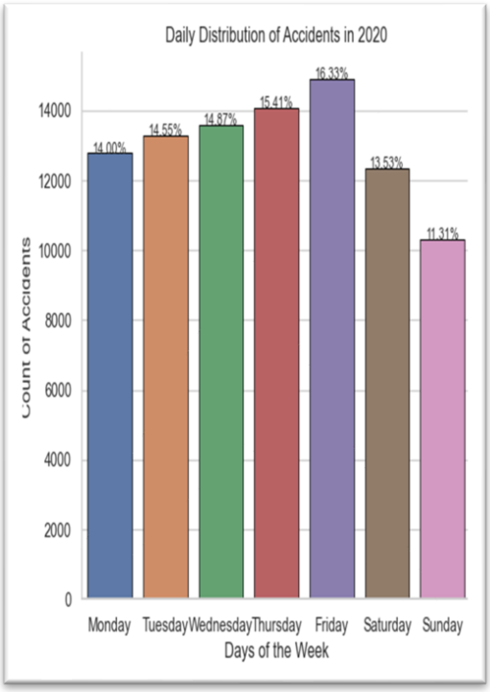


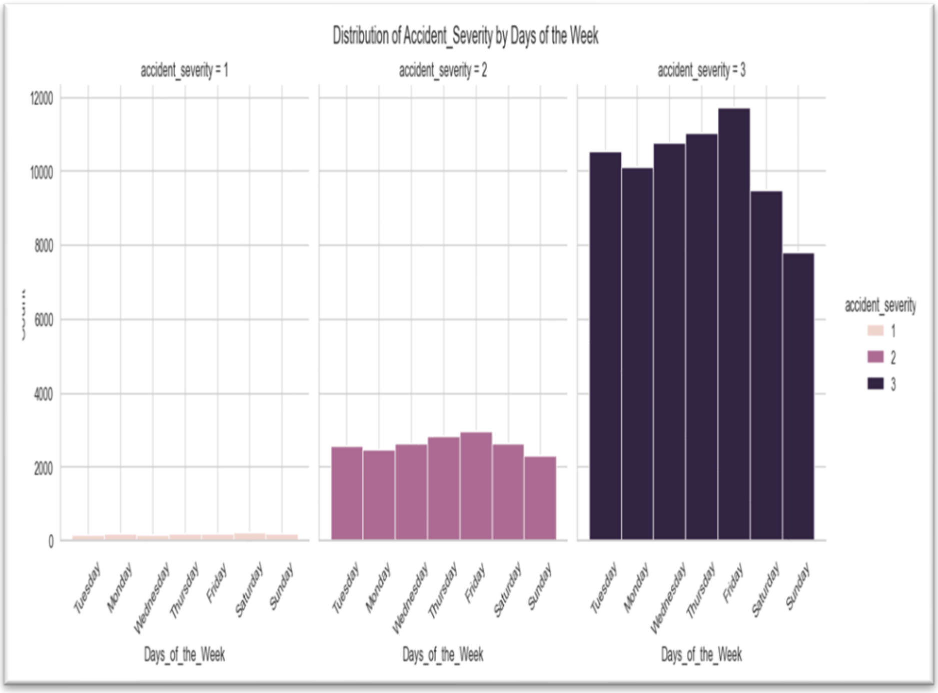

**Accident Distribution Across Hours of the Day and Severity Levels**

| Time | Fatal (1) | Serious (2) | Slight (3) | Total per Hr | Hourly Accidents as a % of Total Accidents | Fatal & Serious Accidents as a % of Hourly Accidents |
|---|---|---|---|---|---|---|
| 0:00 | 40 | 291 | 857 | **1,188** | **1.30%** | **28%** |
| 1:00 | 34 | 236 | 645 | **915** | **1.00%** | **30%** |
| 2:00 | 39 | 170 | 449 | **658** | **0.72%** | **32%** |
| 3:00 | 30 | 129 | 407 | **566** | **0.62%** | **28%** |
| 4:00 | 25 | 125 | 358 | **508** | **0.56%** | **30%** |
| 5:00 | 34 | 219 | 602 | **855** | **0.94%** | **30%** |
| 6:00 | 40 | 409 | 1,381 | **1,830** | **2.01%** | **25%** |
| 7:00 | 35 | 716 | 2,985 | **3,736** | **4.10%** | **20%** |
| 8:00 | 52 | 883 | 4,332 | **5,267** | **5.78%** | **18%** |
| 9:00 | 38 | 710 | 3,169 | **3,917** | **4.30%** | **19%** |
| 10:00 | 77 | 863 | 3,233 | **4,173** | **4.58%** | **23%** |
| 11:00 | 72 | 947 | 3,793 | **4,812** | **5.28%** | **21%** |
| 12:00 | 68 | 1,072 | 4,255 | **5,395** | **5.92%** | **21%** |
| 13:00 | 80 | 1,088 | 4,573 | **5,741** | **6.30%** | **20%** |
| 14:00 | 78 | 1,269 | 4,898 | **6,245** | **6.85%** | **22%** |
| 15:00 | 77 | 1,428 | 5,856 | **7,361** | **8.07%** | **20%** |
| 16:00 | 83 | 1,496 | 5,802 | **7,381** | **8.09%** | **21%** |
| 17:00 | 88 | 1,582 | 6,143 | **7,813** | **8.57%** | **21%** |
| 18:00 | 89 | 1,326 | 5,203 | **6,618** | **7.26%** | **21%** |
| 19:00 | 66 | 1,032 | 3,950 | **5,048** | **5.54%** | **22%** |
| 20:00 | 70 | 749 | 2,896 | **3,715** | **4.07%** | **22%** |
| 21:00 | 69 | 657 | 2,380 | **3,106** | **3.41%** | **23%** |
| 22:00 | 54 | 550 | 1,951 | **2,555** | **2.80%** | **24%** |
| 23:00 | 53 | 408 | 1,335 | **1,796** | **1.97%** | **26%** |
| **TOTAL** | **1,391** | **18,355** | **71,453** | **91,199** | **100%** | **22%** |

### Significant days of the week in which accidents occur

To examine the day of the week with significant number of accidents, two temporary features ('n_time' and 'Days_of_the_Week') was created as a datetime64[ns]object for ease of analysis.



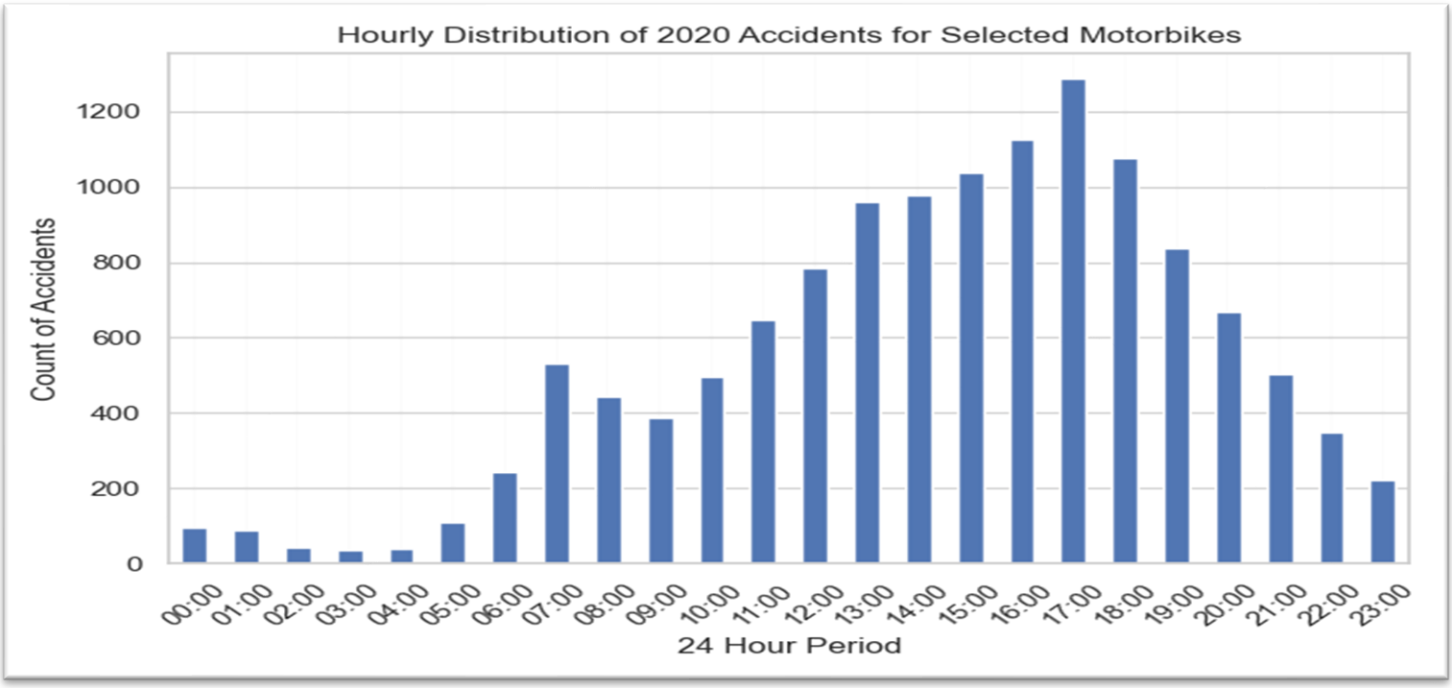



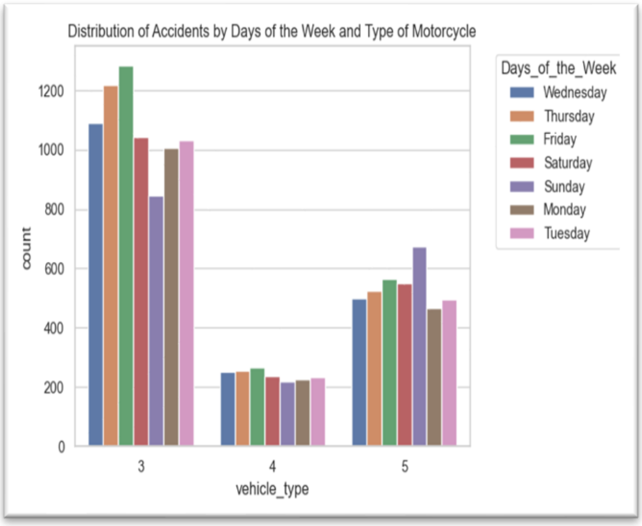
**Interpretation and Analysis for Task No. 1**

- 16.33% of the total accidents in 2020 occurred around 5pm, a count of 7,813 accidents occurrences, followed by 4pm with 7,380 and 3pm with 7,361 incidents. The least number of accidents occurred around 4am with a total number of 508 incidents. This Increased by 68% to 854 by 5am, from 5am it increased by 114% to 1830 at 6am, and a further increment of 104% from 6am to 7am to peak at 8am with a count of 5,267 accidents. It then dropped from 8am to 9am by 25%, before resuming another steady incline from 9am till 5pm. This corresponds with the time the roads are expected to be busy.
- In addition, it is observed that greater number of accidents occur during the daytime, from 8am to 7pm than in the evening from 8pm to 7am. However, as shown by the table of Accident Distribution Across Hours and Severity Levels, a higher percentage of accidents that happen at night time from 8pm to 7am are fatal or serious.
- Friday recorded the highest number of accidents, a total of 14,886, with Sunday recording the lowest number of accidents. The chart showed a steady increase of accidents from Monday peaking at Friday before dropping drastically on Saturday, and flattening on Sunday.
- The third graph 'Distribution of Accident_Severity by Days of the Week' indicates that fatal accidents have the least occurrences, followed by serious accidents and a larger number of slight accidents. This analyses of accidents by severity also reflect the total number of accidents by days of the week.
- In conclusion, most accidents occur between 3pm and 5pm, and most accidents occur during the weekday(workday) than on weekends.

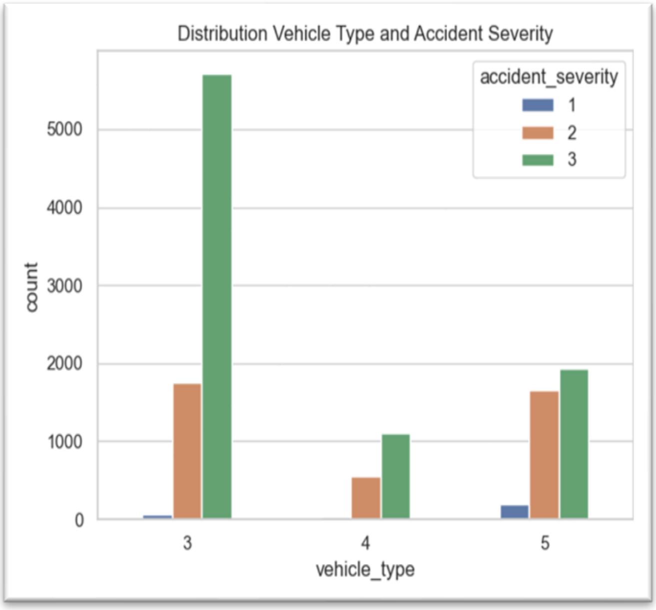
**Task No. 2**
Significant hours of the day, and days of the week, on which accidents occur for selected motorbikes

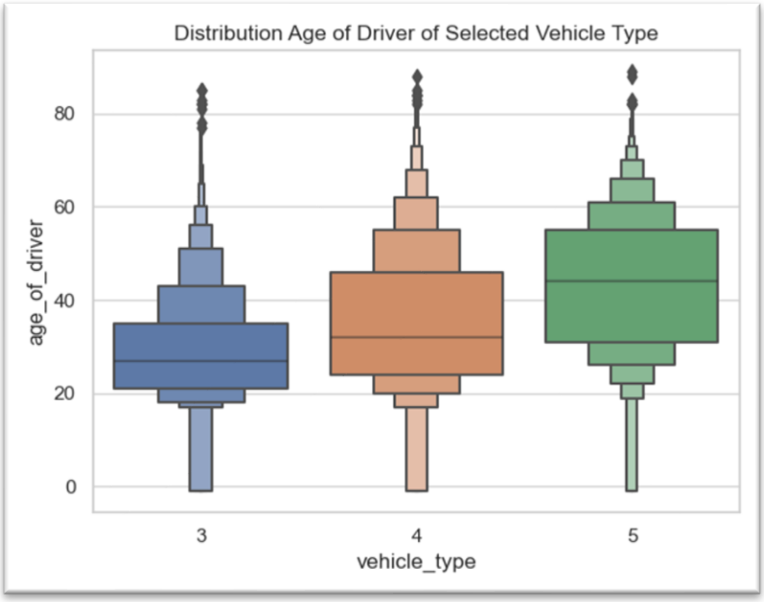
To answer this question, I filtered the Vehicle table then merged it with Accident table.




| Count of Accidents Involving Selected M/Cycles by Days of the Week, Ranked | | | |
|---|---|---|---|
| | **Days** | **Count** | **%** |
| 1 | Friday | 2,119 | 16.30% |
| 2 | Thursday | 2,002 | 15.40% |
| 3 | Wednesday | 1,841 | 14.16% |
| 4 | Saturday | 1,830 | 14.08% |
| 5 | Tuesday | 1,765 | 13.58% |
| 6 | Sunday | 1,738 | 13.37% |
| 7 | Monday | 1,703 | 13.10% |
| | **Total** | **12,998** | **100.00%** |

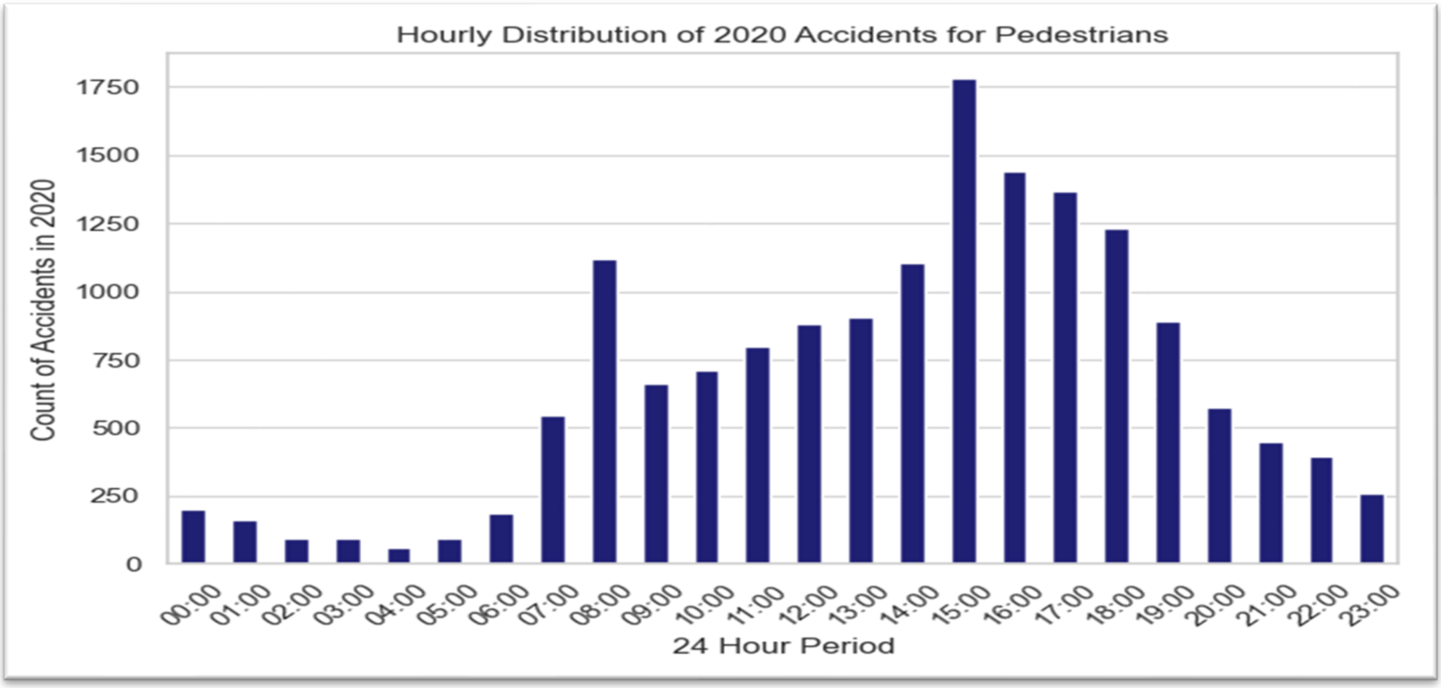


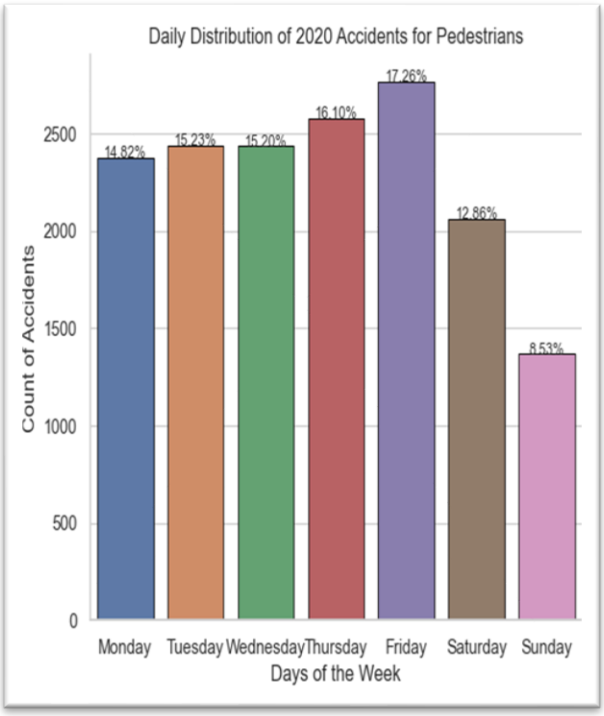



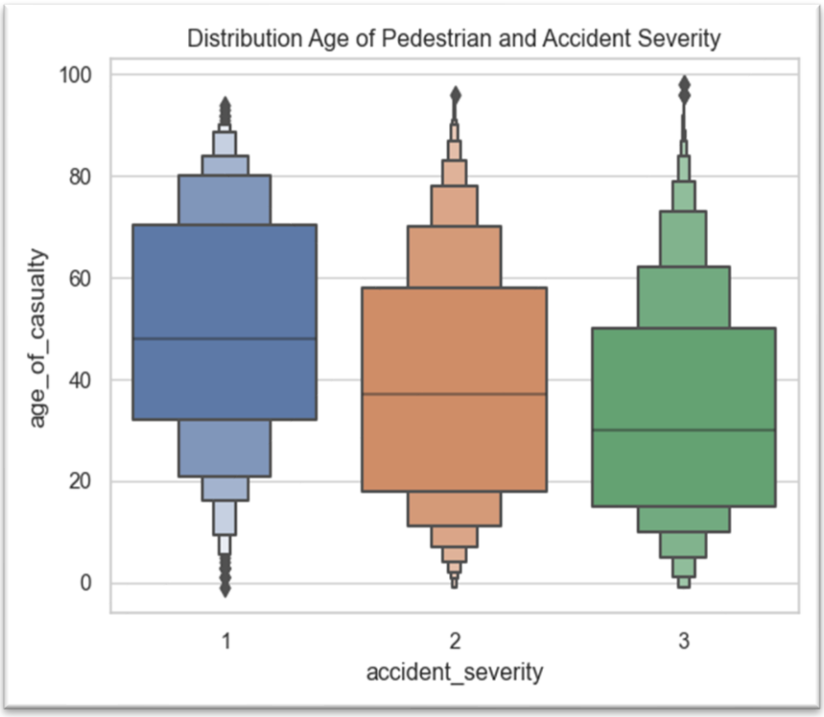

**Interpretation and Analysis for Exercise No. 2**

- From observing the trend from the first bar chart and the complementary table, it is clear that vehicle type 3 (M/cycle 125cc and under) was involved in the highest number of accidents followed by vehicle type 5(M/cycle over 500cc).
- Vehicle type 4 (Motorcycle over 125cc and up to 500cc) was involved in the least number of accidents. This may be as a result of the age bracket of majority of its riders; (25 to 45 years). It can be implied that this age group learnt to balance good experience with judiciously sized motorcycle (note that the higher the cc of a bike, the more difficult it will handle). This vehicle type also recorded the least of fatal accidents. The chart showing the age distribution of vehicles type also illustrates that the smaller the cc of the motorcycle, the younger the rider. This is typical as riders tend to prefer bigger engine as they grow in experience.
- The way count of accidents is distributed across hours is reflective of the total number of accidents in the entire dataset. Most accidents occur around 5pm, with least number of accidents occurring around 2am, same with the entire dataset, there is a spike from 6am to 7am, dropping down gradually rising till it peaked at 5pm.

**Task No. 3**
For pedestrians involved in accidents, are there significant hours of the day, and days of the week, on which they are more likely to be involved?



**Analysis and Interpretation for Task No. 3**

- Similar to exercise 1 and 2, Friday and Sunday have the highest and lowest number of accidents for pedestrians. This reduces the likelihood that these numbers may be biased as a result of a disproportionate contribution by a few or single category. Also, like the previous analysis, the highest number of accidents in the morning occurred around 8am. However, the highest number of accidents for pedestrians occurred around 3pm (this coincides with school closing hour) in contrast to 5pm as observed in exercise no. 1. Overall, the trend is very similar in both exercise 1, 2 and 3, with number of accidents increasing gradually from Monday, reaching its highest point on Friday and dropping to its lowest on Sunday.
- An interesting discovery of the Age of Pedestrian and Accident Severity chart is that older people make up the majority of pedestrians involved in fatal accidents. Age group of pedestrians increased with the severity of accidents, i.e., the older the pedestrian is the more likely they will be in a serious or fatal accident.
- Although the charts and graphs above offer further insights and additional perspective for analysis, I will stop the discussion here, due to word count limitations.

**Task No. 4**

Using the apriori algorithm, explore the impact of selected variables on accident severity.

To report on this question, I first identified accident severity as the target, then select the variables that are much likely to impact on accident severity.

- Some of the variables selected are road_type, speed_limit, light_conditions, weather_conditions, road_surface_conditions, vehicle_type, casualty_class, pedestrian_location

**The Association Rule**

| | antecedents | consequents | antecedent support | consequent support | support | confidence | lift | leverage | conviction | zhangs_metric |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | (road_type) | (accident_severity) | 0.943661 | 0.980806 | 0.924685 | 0.979891 | 0.999067 | -8.636044e-04 | 0.954490 | -0.016307 |
| 1 | (accident_severity) | (road_type) | 0.980806 | 0.943661 | 0.924685 | 0.942781 | 0.999067 | -8.636044e-04 | 0.984612 | -0.046401 |
| 2 | (speed_limit) | (accident_severity) | 0.999936 | 0.980806 | 0.980743 | 0.980805 | 0.999999 | -1.219018e-06 | 0.999936 | -0.019195 |
| 3 | (accident_severity) | (speed_limit) | 0.980806 | 0.999936 | 0.980743 | 0.999935 | 0.999999 | -1.219018e-06 | 0.980806 | -0.000065 |
| 4 | (light_conditions) | (accident_severity) | 0.289396 | 0.980806 | 0.282129 | 0.974888 | 0.993966 | -1.712821e-03 | 0.764316 | -0.008471 |
| 5 | (weather_conditions) | (accident_severity) | 0.219108 | 0.980806 | 0.215161 | 0.981987 | 1.001204 | 2.587818e-04 | 1.065568 | 0.001540 |
| 6 | (road_surface_conditions) | (accident_severity) | 0.311584 | 0.980806 | 0.304466 | 0.977156 | 0.996279 | -1.137244e-03 | 0.840224 | -0.005397 |
| 7 | (vehicle_type) | (accident_severity) | 0.920802 | 0.980806 | 0.902366 | 0.979978 | 0.999156 | -7.625219e-04 | 0.958640 | -0.010557 |
| 8 | (accident_severity) | (vehicle_type) | 0.980806 | 0.920802 | 0.902366 | 0.920025 | 0.999156 | -7.625219e-04 | 0.990279 | -0.042169 |
| 9 | (casualty_class) | (accident_severity) | 0.267675 | 0.980806 | 0.261002 | 0.975070 | 0.994152 | -1.535445e-03 | 0.769908 | -0.007969 |
| 10 | (casualty_type) | (accident_severity) | 0.780198 | 0.980806 | 0.764411 | 0.979765 | 0.998939 | -8.119456e-04 | 0.948569 | -0.004809 |
| 11 | (accident_severity) | (casualty_type) | 0.980806 | 0.780198 | 0.764411 | 0.779370 | 0.998939 | -8.119456e-04 | 0.996248 | -0.052438 |
| 12 | (pedestrian_location) | (accident_severity) | 0.061306 | 0.980806 | 0.059396 | 0.968847 | 0.987807 | -7.331598e-04 | 0.616119 | -0.012979 |
| 13 | (road_type) | (speed_limit) | 0.943661 | 0.999936 | 0.943598 | 0.999933 | 0.999996 | -3.578109e-06 | 0.943661 | -0.000067 |
| 14 | (speed_limit) | (road_type) | 0.999936 | 0.943661 | 0.943598 | 0.943658 | 0.999996 | -3.578109e-06 | 0.999936 | -0.056342 |
| 15 | (light_conditions) | (road_type) | 0.289396 | 0.943661 | 0.272865 | 0.942878 | 0.999170 | -2.267856e-04 | 0.986281 | -0.001168 |

**Analysis and Interpretation**

Notice that the tables have very high confidence values, meaning that the accident is frequently found where the various selected variables (antecedents) like speed_limit, road_type, whether_conditions,etc is present. It is also evident that support (a metric that measures events that frequently occur together) is high in most of the observations, except in areas like pedestrian_location vs accident_severity and some few others.

In contrast, most values under lift are 1 or close to 1, there are no lift values that are more than 1. Note that lift measures how much likely a Consequence will appear when the Antecedent is present. Lift values that is greater than 1 indicates a strong association between the Consequence and the Antecedent, Lift values that are equal to 1 means that there is no relationship between the 2 events under consideration. Lift values less than 1 means a negative relationship, the occurrence of one event indicates the non-occurrence of the other. Similarly, Zhangs metric, an extension of the lift metric that measures the association/dissociation between variables, is mostly negative. A negative Zhang metric (-1: perfect dissociation) indicates lack of association while a positive Zhang metric indicates association (Bismi, 2023).

Despite the high values for Support and Confidence suggesting that there is a relationship between accident_severity and the selected variables, lift and Zhangs _metric indicate the opposite. This suggests that although the presence of the selected variables influence the occurrence of accident, it does not necessarily influence the severity of the accidents that occurred.

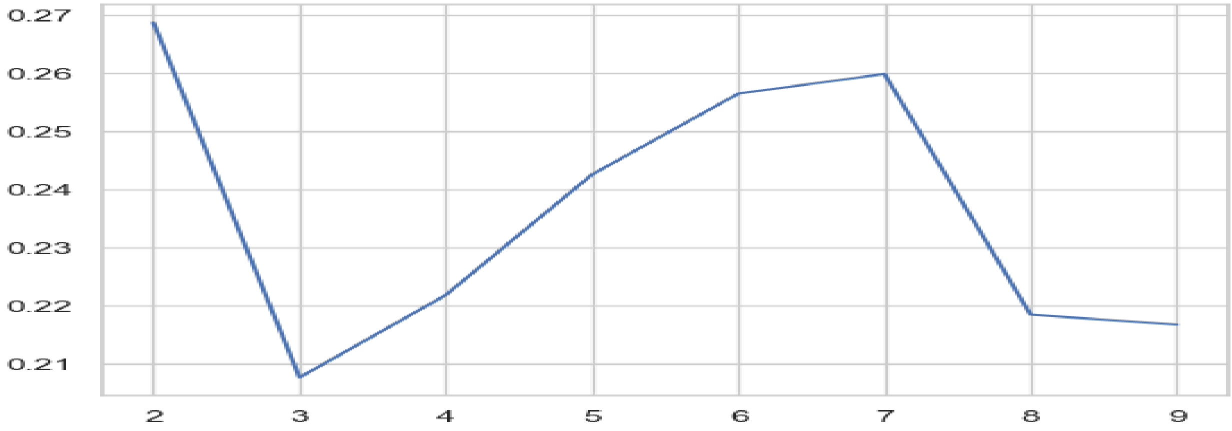
**Task No. 5**

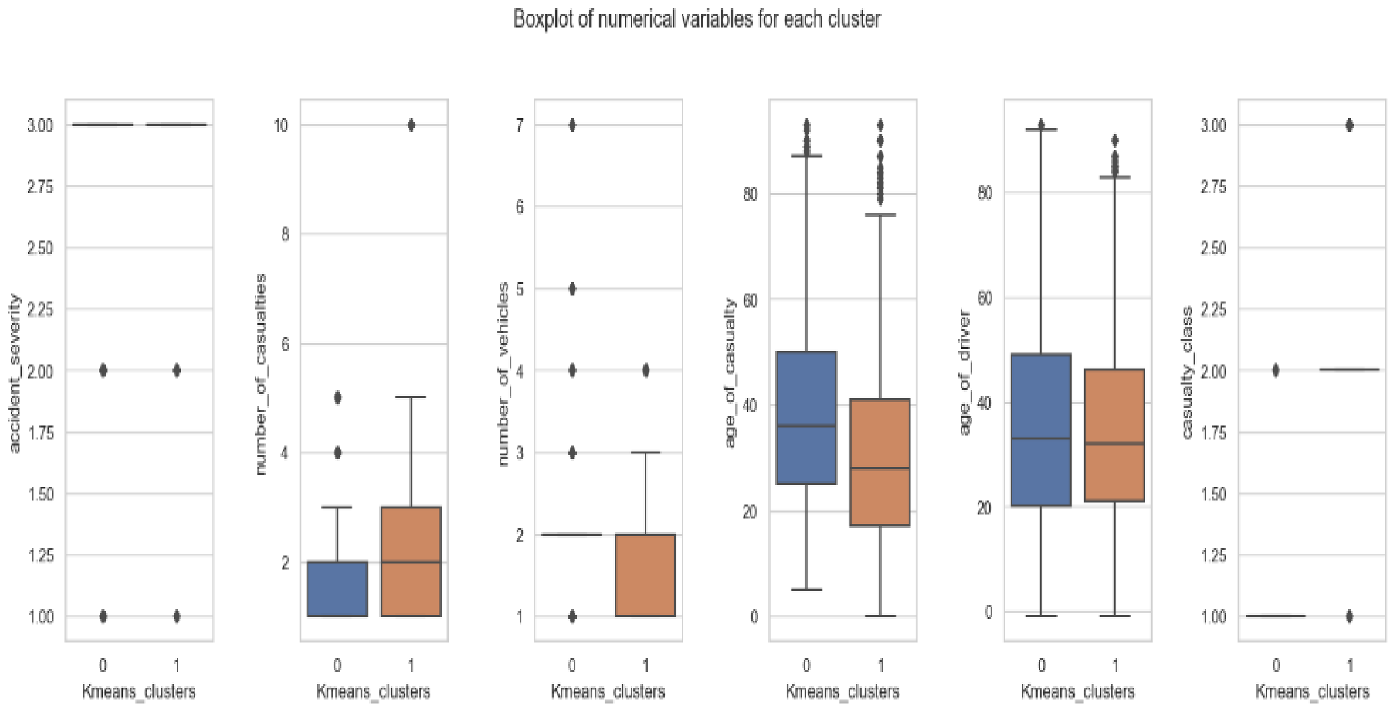
Identify accidents in our region. Run clustering on this data. What do these clusters reveal about the distribution of accidents in our region?

I merged the four tables then filter for unitary authorities (North Lincolnshire, North East Lincolnshire, Kingston upon Hull and East Riding of Yorkshire) in Humberside using the lsoa1nm column. Then use the KMeans clustering on selected variables, based on calculated optimal number of clusters,

**Calculation of optimal number of clusters**

```
At n_clusters == 2, silhouette score is 0.2687027379759941
At n_clusters == 3, silhouette score is 0.20765690691985306
At n_clusters == 4, silhouette score is 0.22182596896965737
At n_clusters == 5, silhouette score is 0.24254033268930664
At n_clusters == 6, silhouette score is 0.25644458773077095
At n_clusters == 7, silhouette score is 0.25982863433019826
At n_clusters == 8, silhouette score is 0.21845335999735663
At n_clusters == 9, silhouette score is 0.21673912522986388
```





**Analysis and Interpretation for Task 5**

Based on the optimal number of clusters as calculated, the accidents in Humberside were clustered into 2 groups. There are some characteristics that are prevalent in the entire dataset therefore they are shared by the two groups. For example, the two groups mostly consist of slight accidents, with few that are serious and fatal. Other example are the age of casualty and age of driver. However, the groups differed on number of vehicles and casualty class. The first group has mainly casualty_class 1 (drivers/riders) as the majority of the casualties, with no pedestrian casualty in the group, while the second group is more of passengers with some drivers and pedestrians in the group too.

**Exercise No. 6**

Using outlier detection methods, identify unusual entries in your dataset. Should you keep these entries in your data?

Two model-based outlier detection methods were used on the dataset filtered for Humber Region.

### I. Isolation Forest

Isolation Forest detected the following numbers of outliers in the various columns in the dataset.

| | Features | No. of Outliers |
|---|---|---|
| 1 | police_force | 4,464 |
| 2 | number_of_vehicles | 512 |
| 3 | number_of_casualties | 476 |
| 4 | road_type | 1,417 |
| 5 | speed_limit | 11,540 |
| 6 | light_conditions | 791 |
| 7 | weather_conditions | 750 |
| 8 | road_surface_conditions | 419 |
| 9 | vehicle_type | 8,785 |
| 10 | sex_of_driver | 478 |
| 11 | age_of_driver | 8,931 |
| 12 | engine_capacity_cc | 471,037 |
| 13 | age_of_vehicle | 1,500 |
| 14 | casualty_class | 581 |
| 15 | sex_of_casualty | 404 |
| 16 | age_of_casualty | 11,333 |
| 17 | casualty_severity | 707 |
| 18 | pedestrian_location | 795 |
| 19 | Our_Region_East Riding of Yorkshire | 100 |
| 20 | Our_Region_Kingston upon Hull | 48 |
| 21 | Our_Region_North East Lincolnshire | 49 |
| 22 | Our_Region_North Lincolnshire | 82 |

### II. Local Outlier Factor

LOC detected the following outliers in the data

| | Features | No. of Outliers |
|---|---|---|
| 1 | police_force | 1,632 |
| 2 | number_of_vehicles | 224 |
| 3 | number_of_casualties | 122 |
| 4 | road_type | 585 |
| 5 | speed_limit | 3,370 |
| 6 | light_conditions | 151 |
| 7 | weather_conditions | 146 |
| 8 | road_surface_conditions | 125 |
| 9 | vehicle_type | 795 |
| 10 | sex_of_driver | 132 |
| 11 | age_of_driver | 2,213 |
| 12 | engine_capacity_cc | 130,781 |
| 13 | age_of_vehicle | 532 |
| 14 | casualty_class | 123 |
| 15 | sex_of_casualty | 110 |
| 16 | age_of_casualty | 2,824 |
| 17 | casualty_severity | 282 |
| 18 | pedestrian_location | 24 |
| 19 | Our_Region_East Riding of Yorkshire | 33 |
| 20 | Our_Region_Kingston upon Hull | 45 |
| 21 | Our_Region_North East Lincolnshire | 15 |
| 22 | Our_Region_North Lincolnshire | 9 |

**I think these outliers should be kept and the following are my reasons.**

- In this dataset negative values are used to denote unknown values and unknown values are normal part of data collection.
- Deleting a critical dataset like the Accident will definitely have socio-political and legal and consequences.
- Outliers are rare by definition. If there are many of them, that means there is something the outliers are saying and they need to be looked further into (Ferguson, 2019).

**Exercise No. 7**
Can you develop a classification model using the provided data that accurately predicts fatal injuries sustained in road traffic accidents, with the aim of informing and improving road safety measures?

**Gradient Boost Classifier Model**

The model is to predict fatal injuries sustained in road traffic accidents. Using Gradient Boosting method

The classification report after the model was balanced using SMOTE

Classification Report2:

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| Fatal | 0.98 | 0.65 | 0.78 | 683 |
| Serious | 0.97 | 0.75 | 0.85 | 6361 |
| Slight | 0.93 | 1.00 | 0.96 | 23340 |
| accuracy | | | 0.94 | 30384 |
| macro avg | 0.96 | 0.80 | 0.86 | 30384 |
| weighted avg | 0.94 | 0.94 | 0.94 | 30384 |

**Analysis and Interpretation**

The model has a high precision, which means that out of the total number of predicted fatal accidents, the model was able to predict 98% correctly. Recall of 65% means that out of all fatal accidents in the test data, the model was able to predict 65% of them correctly. A high accuracy is also good because the model was balanced using SMOTE The F1 score is also good, it is moderate (not high, not low), it is a balanced measure between precision and recall.Support of 683 means that there are 683 instances of fatal accident

**Recommendations.**

Recommendations to government agencies based to improve safety.

- There should be more traffic officials in the morning and late afternoon especially on Fridays, to minimise the number of accidents, particularly those occasioned as a result of breaching traffic laws.
- It was established that pedestrians are involved in most in accidents around 3pm in contrast to vehicular accident which occurred the most at around 5pm. This coincides with school closing period. And it can be prevented by encouraging pedestrians to us side-walks and cross walks, and designated walk lanes. Awareness should be raised especially in schools.
- The boxenplot of age_of_drivers vs casualty_class indicates that are a good number of drivers who are underaged. A severer penalty should be imposed on under drivers to discourage under age driving. The plot also shows that there are more underage drivers involved in pedestrian accidents than any other class of accidents.
- Vehicle type 3 was involved a far greater number of accidents that the other 2 types, though most of these are of slight severity. This shows that most riders maybe self-taught and first-time riders. Government may develop a special policy to cater to these first-time riders.

**References**

Bismi, I. (2023) How to perform market basket analysis using Apriori Algorithm and Association rules, Medium. Available at: https://medium.com/@iqra.bismi/how-to-perform-market-basket-analysis-using-apriori-algorithm-and-association-rules- (Accessed: 08 August 2023).

Ferguson, K. (2019) *When should you delete outliers from a data set? - atlan: Humans of data*, *Atlan*. Available at: https://humansofdata.atlan.com/2018/03/when-delete-outliers-dataset/ (Accessed: 12 August 2023).

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